

**Multiple Choice:**

1. A man standing on a 48.5 meter building high, has an eyesight height of 1.5 m from the top of the building, took a depression reading from the top of another nearby building and nearest wall, which are 50° and 80° respectively. Find the height of the nearby building in meters. The man is standing at the edge of the building and both buildings lie on the same horizontal plane.
A. 39.49
B. 35.50
C. 30.74
D. 42.55
2. $\sin(B - A)$ is equal to _____, when $B = 270$ degrees and A is an acute angle.
A. $-\cos A$
B. $\cos A$
C. $-\sin A$
D. $\sin A$
3. Points A and B are 100 m apart and are of the same elevation as the foot of a building. The angles of elevation of the top of the building from points A and B are 21° and 32° respectively. How far is A from the building in meters?
A. 259.28
B. 265.42
C. 271.64
D. 277.29
4. An observer wishes to determine the height of a tower. He takes sights at the top of the tower from A and B, which are 50 feet apart, at the same elevation on a direct line with the tower. The vertical angle at point A is 30° and at point B is 40° . What is the height of the tower?
A. 85.60 feet
B. 92.54 feet
C. 110.29 feet
D. 143.97 feet
5. The sides of a triangular lot are 130 m., 180 m and 190 m. The lot is to be divided by a line bisecting the longest side and drawn from the opposite vertex. Find the length of the line.
A. 120 m
B. 130 m
C. 125 m
D. 128 m
6. The sides of a triangle are 8, 15 and 17 units. If each side is doubled, how many square units will the area of the new triangle be?
A. 240
B. 420
C. 320
D. 200
7. Two triangles have equal bases. The altitude of one triangle is 3 units more than its base and the altitude of the other triangle is 3 units less than its base. Find the altitudes, if the areas of the triangles differ by 21 square units.
A. 6 and 12
B. 3 and 9
C. 5 and 11
D. 4 and 10
8. From a point outside of an equilateral triangle, the distances to the vertices are 10 m, 18 m and 10 m, respectively. What is the length of one side of a triangle?
A. 17.75 m
B. 18.50 m
C. 19.95 m
D. 20.50 m
9. The sides of a triangle are 8 cm, 10 cm and 14 cm. Determine the radius of the inscribed circle.
A. 2.25 cm
B. 2.35 cm
C. 2.45 cm
D. 2.55 cm
10. The point of concurrency of the angle bisector of the triangle is called _____.
A. orthocenter
B. circumcenter
C. centroid
D. incenter
11. What is the radius of the circumscribed circle of a right triangle?
A. hypotenuse/3
B. hypotenuse/2
C. hypotenuse
D. product of perpendicular sides/perimeter of triangle
12. In triangle ABC, $AB = 13$, $CB = 15$ and $CA = 14$. Point D is on BC with $CD = 6$. Point E is on BC such that $\angle BAE \cong \angle CAD$. Find BE.
A. 5.47
B. 4.46
C. 6.12
D. 7.83
13. The angle of depression of a ship viewed at a particular instant from the top of a 75 m vertical cliff is 30° . The ship is sailing away from the cliff at constant speed and 1 minute later its angle of depression from the top of the cliff is 20° . Determine the speed of the ship in km/h.
A. 4.21
B. 5.12
C. 4.57
D. 5.76
14. A point P within an equilateral triangle has distance of 4, 5 and 6 units respectively from the vertices. Determine the length of the sides of the equilateral triangle.
A. 9.50 units
B. 7.55 units
C. 8.53 units
D. 9.12 units
15. A paper equilateral triangle ABC has side length 12. The paper triangle is folded so that vertex A touches a point on side BC a distance 9 from point B. Find the length of the line segment along which the triangle is folded.



- A. 5.67
B. 6.96
C. 6.34
D. 5.12
16. The area of a triangle is 8346 sq. m and two of its interior angles are $37^\circ 25'$ and $56^\circ 17'$. What is the length of the longest side?
- A. 171.5 m
B. 181.5 m
C. 191.5 m
D. 200.5 m
17. In triangle ABC, angle $C = 70^\circ$, $A = 45^\circ$, $AB = 40$ m. What is the length of the median drawn from vertex A to side BC?
- A. 36.3 m
B. 36.6 m
C. 36.9 m
D. 37.2 m
18. Simplify the following:
$$\frac{\cos A + \cos B}{\sin A - \sin B} + \frac{\sin A + \sin B}{\cos A - \cos B}$$
- A. 0
B. $\sin A$
C. 1
D. $\cos A$
19. An airplane can fly at an air speed of 300 mph. If there is a wind blowing towards the east at 50 mph, what should be the plane's compass heading in order for its course to be 30° ? What will be the plane's ground speed if it flies in this course?
- A. 19.7° , 307.4 mph
B. 20.1° , 309.4 mph
C. 21.7° , 321.8 mph
D. 22.3° , 319.2 mph
20. The sides of a triangle are 195, 157 and 210, respectively. What is the area of the triangle?
- A. 73,250 sq. units
B. 10,250 sq. units
C. 14,586 sq. units
D. 11,260 sq. units
21. A pole cast a shadow 15 m long when the angle of elevation of the sun is 61° . If the pole is leaned 15° from the vertical directly towards the sun, determine the length of the pole.
- A. 54.23 m
B. 48.23 m
C. 42.44 m
D. 46.21 m
22. If an equilateral triangle is circumscribed about a circle of radius 10 cm, determine the side of the triangle.
- A. 34.64 cm
B. 64.12 cm
C. 36.44 cm
D. 32.10 cm
23. Points A and B 1000 m apart are plotted on a straight highway running East and West. From A, the bearing of a tower C is 32° W of N and from B the bearing of C is 26° N of E. Approximate the shortest distance of tower C to the highway.
- A. 364 m
B. 374 m
C. 384 m
D. 394 m
24. The angle of elevation of the top of tower B from the top of tower A is 28° and the angle of elevation of the top of tower A from the base of tower B is 46° . The two towers lie in the same horizontal plane. If the height of tower B is 120 m, find the height of tower A.
- A. 66.3 m
B. 79.3 m
C. 87.2 m
D. 90.7 m
25. The captain of a ship views the top of a lighthouse at an angle of 60° with the horizontal at an elevation of 6 meters above sea level. Five minutes later, the same captain of the ship views the top of the same lighthouse at an angle of 30° with the horizontal. Determine the speed of the ship if the lighthouse is known to be 50 meters above sea level.
- A. 0.265 m/sec
B. 0.155 m/sec
C. 0.169 m/sec
D. 0.210 m/sec
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